

Mini Project Report

Air Quality Index (AQI) Monitoring

Submitted By

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CSE Department | Section S5

R-Programming Lab

SCSVMV UNIVERSITY

Experiment 3: Air Quality Index (AQI) Monitoring

Aim

To collect, analyse, and visualize Air Quality Index (AQI) data from multiple cities using R programming, and to interpret pollution levels using bar charts, histograms, box plots, and pie charts.

Scenario / Background

- Environmental monitoring agencies collect daily AQI readings from sensors placed across major Indian cities.
- The AQI value reflects the concentration of key pollutants: PM2.5, PM10, CO, NO2, SO2, and O3.
- Government bodies and the public use AQI categories (Good, Moderate, Unhealthy, Very Unhealthy, Hazardous) to decide health advisories.
- Analysing AQI trends helps identify cities with chronic pollution problems and the most common primary pollutants.
- Visualizing the data allows policymakers to take targeted action to improve air quality and public health outcomes.

Sample Dataset

The dataset contains 10 records from 5 Indian cities collected over 10 consecutive days.

Station ID	Date	City	AQI Value	Pollutant Conc. (µg/m³)	Primary Pollutant	AQI Category
S001	2024-01-15	Chennai	142	45	PM2.5	Unhealthy
S002	2024-01-16	Mumbai	89	30	PM10	Moderate
S003	2024-01-17	Delhi	210	60	PM2.5	Very Unhealthy
S004	2024-01-18	Kolkata	55	20	CO	Good
S005	2024-01-19	Bangalore	78	25	NO2	Moderate
S006	2024-01-20	Chennai	165	50	PM2.5	Unhealthy
S007	2024-01-21	Mumbai	38	10	SO2	Good
S008	2024-01-22	Delhi	295	80	PM2.5	Hazardous
S009	2024-01-23	Kolkata	110	35	PM10	Unhealthy for Sensitive
S010	2024-01-24	Bangalore	62	22	O3	Moderate

R Program (with line-by-line comments)

```

# =====
# Experiment 3: Air Quality Index Monitoring
# =====

# Step 1: Load required libraries
library(ggplot2) # For creating visualizations
library(dplyr)   # For data manipulation

# Step 2: Define the AQI dataset inline
aqi_data <- data.frame(
  StationID = c("S001", "S002", "S003", "S004", "S005",
    "S006", "S007", "S008", "S009", "S010"),
  Date      = as.Date(c("2024-01-15", "2024-01-16", "2024-01-17",
    "2024-01-18", "2024-01-19", "2024-01-20",
    "2024-01-21", "2024-01-22", "2024-01-23",
    "2024-01-24")),
  City      = c("Chennai", "Mumbai", "Delhi", "Kolkata", "Bangalore",
    "Chennai", "Mumbai", "Delhi", "Kolkata", "Bangalore"),
  AQI       = c(142, 89, 210, 55, 78, 165, 38, 295, 110, 62),
  Pollutant_Conc = c(45, 30, 60, 20, 25, 50, 10, 80, 35, 22),
  Primary_Pollutant = c("PM2.5", "PM10", "PM2.5", "CO", "NO2",
    "PM2.5", "SO2", "PM2.5", "PM10", "O3"),
  Category   = c("Unhealthy", "Moderate", "Very Unhealthy", "Good",
    "Moderate", "Unhealthy", "Good", "Hazardous",
    "Unhealthy for Sensitive", "Moderate")
)

# Step 3: View the dataset
View(aqi_data)
cat("AQI Dataset Loaded Successfully\n")
print(summary(aqi_data))

# Step 4: Bar Chart - Average AQI by City
avg_aqi <- aqi_data %>%
  group_by(City) %>%
  summarise(Avg_AQI = mean(AQI)) # Compute mean AQI per city

ggplot(avg_aqi, aes(x = City, y = Avg_AQI, fill = City)) +
  geom_bar(stat = "identity") + # Draw bars with actual values
  labs(title = "Average AQI by City",
    x = "City", y = "Average AQI") +
  theme_minimal() +
  theme(legend.position = "none") # Remove legend (redundant with x-axis)

# Step 5: Histogram - AQI Value Distribution
ggplot(aqi_data, aes(x = AQI)) +
  geom_histogram(binwidth = 40, fill = "steelblue", color = "black") +
  labs(title = "Distribution of AQI Values",
    x = "AQI", y = "Frequency") +
  theme_minimal()

# Step 6: Box Plot - AQI spread per City
ggplot(aqi_data, aes(x = City, y = AQI, fill = City)) +
  geom_boxplot() + # Show median, IQR, and outliers
  labs(title = "AQI Box Plot by City",
    x = "City", y = "AQI") +
  theme_minimal() +
  theme(legend.position = "none")

```

```
# Step 7: Pie Chart - Proportion of AQI Categories
category_counts <- table(aqi_data$Category) # Count each category
pie(category_counts,
     main = "AQI Category Distribution",
     col = rainbow(length(category_counts))) # Assign distinct colors
```

Output / Observation

- Bar Chart: Delhi records the highest average AQI (~252), followed by Chennai (~153), indicating severe pollution levels. Bangalore and Kolkata show relatively lower average AQI values.
- Histogram: The AQI distribution is right-skewed, with most readings clustered between 38 and 165. One extreme outlier (AQI = 295 for Delhi) highlights hazardous episodes.
- Box Plot: Delhi shows the widest interquartile range and highest median AQI, confirming persistent air quality issues. Mumbai and Bangalore have compact, lower-range distributions.
- Pie Chart: PM2.5 is the dominant primary pollutant (~40% of records), followed by PM10 and other pollutants, reflecting the major role of particulate matter in urban air pollution.
- Overall Insight: The analysis reveals that metro cities like Delhi face chronic high-AQI conditions, primarily driven by PM2.5. Smaller cities like Bangalore maintain relatively better air quality. Monitoring such trends is critical for data-driven environmental policymaking.

Graphs Representation

The following graphs are generated by executing the above R program:

- Graph 1: Bar Chart – Average AQI by City
- Graph 2: Histogram – Distribution of AQI Values
- Graph 3: Box Plot – AQI Spread per City
- Graph 4: Pie Chart – AQI Category Distribution

The screenshot displays a Jupyter Notebook environment with the following details:

- Header:** Shows the file name '11249A040.ipynb', a star icon, and a share button. The menu bar includes 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'.
- Toolbar:** Features a search bar, '+ Code', '+ Text', and a 'Run all' button. On the right, it shows 'RAM' and 'Disk' usage indicators.
- Project Title:** 'Air Quality Index Monitoring' is displayed above the code cell.
- Code Cell [1]:** Contains R code for loading libraries, defining the AQI dataset, and setting the date range.


```
# =====
# Project title : Air Quality Index Monitoring
# =====

# Step 1: Load required libraries

library(ggplot2) # For creating visualizations
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# Step 2: Define the AQI dataset inline
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  Date      = as.Date(c("2024-01-15", "2024-01-16", "2024-01-17",
                       "2024-01-18", "2024-01-19", "2024-01-20",
                       "2024-01-21", "2024-01-22", "2024-01-23"))
```



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```
'-
StationID = c("S001", "S002", "S003", "S004", "S005",
"S006", "S007", "S008", "S009", "S010"),
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"2024-01-21", "2024-01-22", "2024-01-23",
"2024-01-24")),
City      = c("Chennai", "Mumbai", "Delhi", "Kolkata", "Bangalore",
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AQI       = c(142, 89, 210, 55, 78, 165, 38, 295, 110, 62),
Pollutant_Conc = c(45, 30, 60, 20, 25, 50, 10, 80, 35, 22),
Primary_Pollutant = c("PM2.5", "PM10", "PM2.5", "CO", "NO2",
"PM2.5", "SO2", "PM2.5", "PM10", "O3"),
Category  = c("Unhealthy", "Moderate", "Very Unhealthy", "Good",
"Moderate", "Unhealthy", "Good", "Hazardous",
"Unhealthy for Sensitive", "Moderate")
)

# Step 3: View the dataset
View(aqi_data)
```



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# Step 7: Pie Chart - Proportion of AQI Categories
category_counts <- table(aqi_data$Category) # Count each category
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```



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...
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
filter, lag

The following objects are masked from 'package:base':
intersect, setdiff, setequal, union

A data.frame: 10 × 7

StationID	Date	City	AQI	Pollutant_Conc	Primary_Pollutant	Category
<chr>	<date>	<chr>	<dbl>	<dbl>	<chr>	<chr>

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StationID	Date	City	AQI	Pollutant_Conc	Primary_Pollutant	Category
<chr>	<date>	<chr>	<dbl>	<dbl>	<chr>	<chr>
S001	2024-01-15	Chennai	142	45	PM2.5	Unhealthy
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S006	2024-01-20	Chennai	165	50	PM2.5	Unhealthy
S007	2024-01-21	Mumbai	38	10	SO2	Good
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S008	2024-01-22	Delhi	295	80	PM2.5	Hazardous
S009	2024-01-23	Kolkata	110	35	PM10	Unhealthy for Sensitive
S010	2024-01-24	Bangalore	62	22	O3	Moderate

AQI Dataset Loaded Successfully

StationID	Date	City	AQI
Length:10	Min. :2024-01-15	Length:10	Min. : 38.0
Class :character	1st Qu.:2024-01-17	Class :character	1st Qu.: 66.0
Mode :character	Median :2024-01-19	Mode :character	Median : 99.5
	Mean :2024-01-19		Mean :124.4
	3rd Qu.:2024-01-21		3rd Qu.:159.2
	Max. :2024-01-24		Max. :295.0

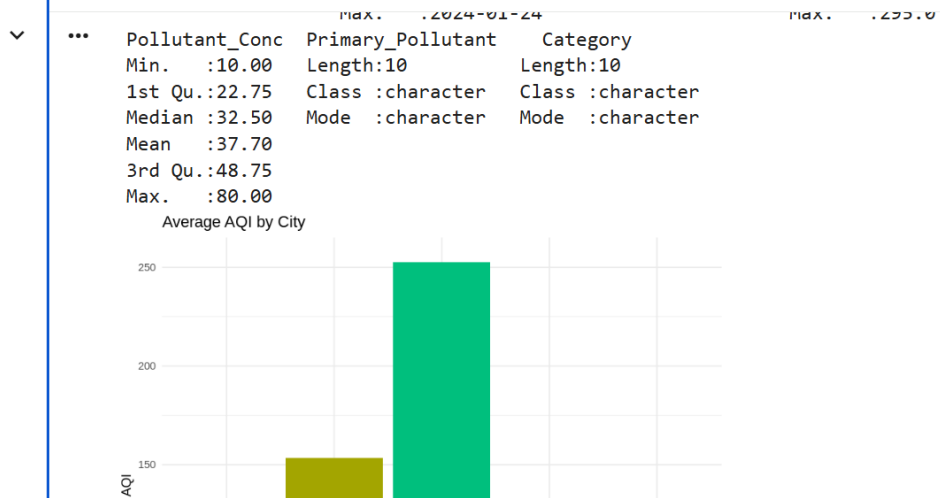
Pollutant_Conc	Primary_Pollutant	Category
Min. :10.00	Length:10	Length:10
1st Qu.:22.75	Class :character	Class :character
Median :32.50	Mode :character	Mode :character
Mean :37.70		



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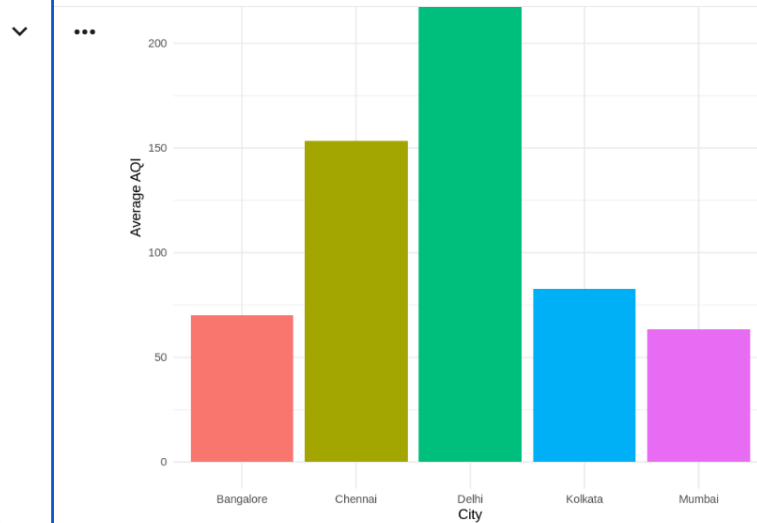
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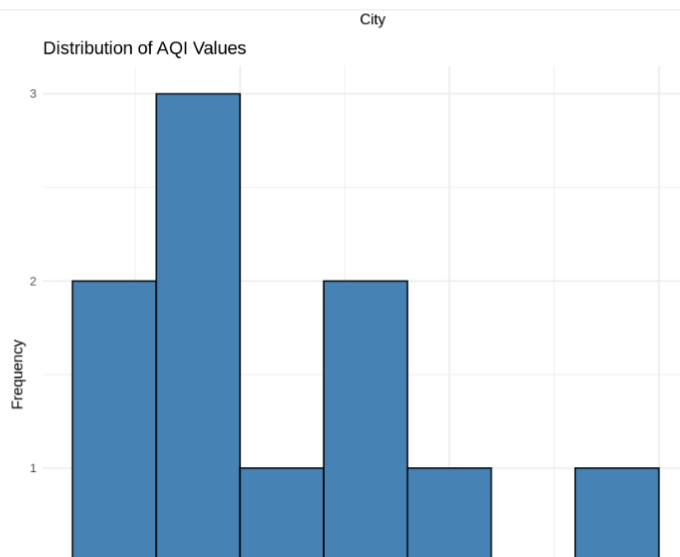




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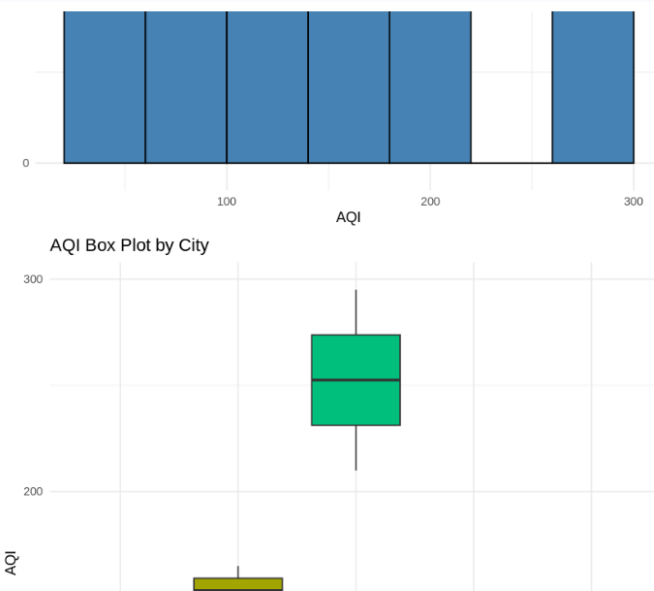
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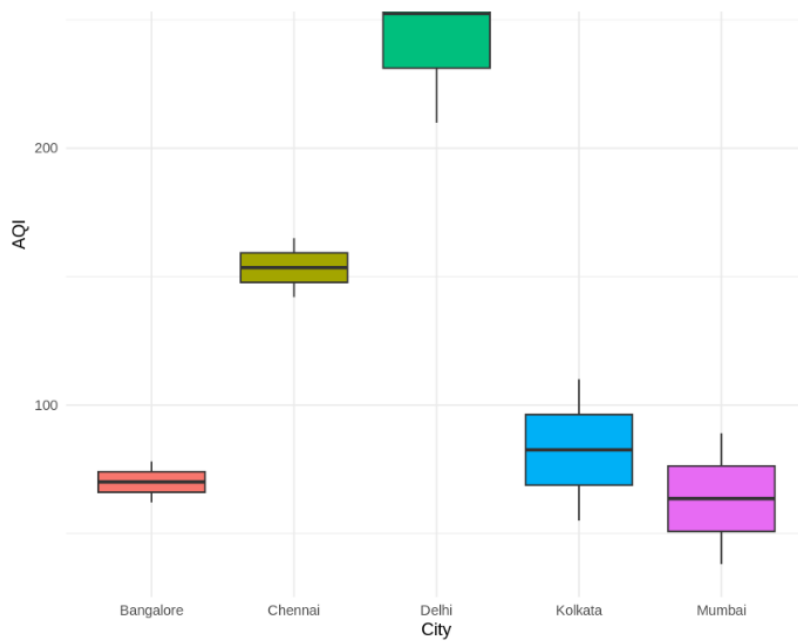
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AQI Category Distribution



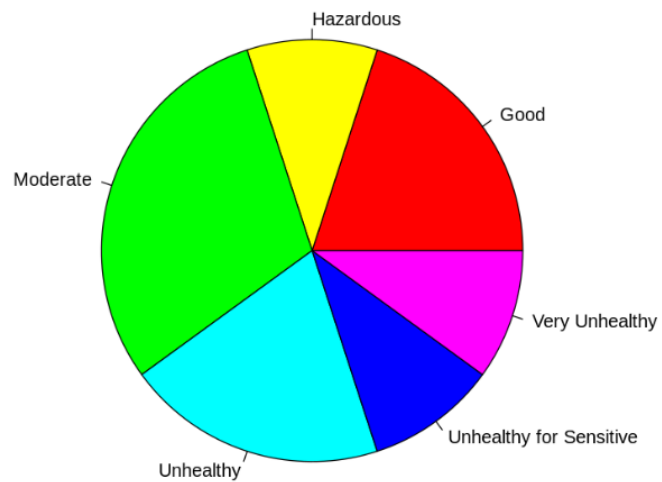
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AQI Category Distribution



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